CONVECTIVE HEAT TRANSFER DUE TO AN IMPINGING SYNTHETIC JET: A NUMERICAL INVESTIGATION OF A CANONICAL GEOMETRY

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ABSTRACT

In the present study, a canonical geometry has been created for investigating the flow and heat transfer of a purely oscillatory jet that is not affected by the manner in which it is produced, and hence has allowed us to investigate the flow and heat transfer in a generalized approach. The unsteady Navier-Stokes equations and the convection-diffusion equation were solved using a full unsteady, two-dimensional finite volume approach in order to capture the complex time dependent flow field produced by periodic ejection-suction at high frequencies. The effect of the Reynolds number *Re* and the jet-to-surface distance *H* upon the heat transfer were parametrically examined, over a range of oscillation frequencies. Results were compared with a steady, laminar impinging jet of the same configuration, and over the same range, for matched Reynolds numbers $Re < 600$. It was found that synthetic and continuous jets are comparable in cooling efficiency, but have dissimilar dependence on jet-to-surface distance. Furthermore, an optimum jet-to-surface distance was indentified for oscillatory jets that does not exist for steady laminar jets.

KEY WORDS: Impinging synthetic jet, Convective heat transfer, Computational fluid dynamics.

NOMENCLATURE

c_p	specific heat, J/kg K
f	frequency, Hz
h	heat transfer coefficient, W/m^2 K
H	jet-to-surface distance, m
k	thermal conductivity, W/m K
L_0	stroke length, m
$\dot{m}$	mass flux, kg/s
Nu	Nusselt number
Nu_{avg}	time averaged Nusselt number
$\overline{Nu}_{avg}$	space and time averaged Nusselt number
p	pressure, Pa
P	pumping power, W
q''	heat flux per unit area, W/m^2
Re	Reynolds number
t	time, s
T	period $1/f$, s
U_0	time and space averaged velocity, m/s
v_{in}	inlet velocity, m/s
v_{max}	inlet maximum velocity, m/s
$\mathbf{v}$	velocity vector, m/s
w	channel width, m
x, y	horizontal and vertical position, m

Greek symbols

θ	temperature, K
θ_{wall}	wall temperature, K
θ_∞	ambient temperature, K
μ	dynamic viscosity, Pa s
ν	kinematic viscosity, m^2/s
ρ	density, kg/m^3

Subscripts

avg	time averaged
in	inlet condition
wall	related to wall
∞	ambient condition

INTRODUCTION

Synthetic jets are created by oscillatory pumping of fluid to and from an orifice. In its most common rendition, they have been fabricated using piezo-resistive actuators to produce periodic volume change in the cavity feeding the orifice, thereby periodically ingesting fluid and ejecting it as a jet, or a puff, into an open plenum. Although such a system has zero net flow from the orifice, it produces a jet with net momentum because the flow patterns are quite distinct in the ingestion compared to the ejection parts of the cycle. As such this jet can be used as a cooling device by configuring it to impinge onto the heated surface. Except for the earliest seminal work on the subject, most of the studies have been directed towards understanding synthetic jets created by pumps fashioned with piezo-resistive actuators as this type of device has shown promise as a small scale cooling technique for air-cooled electronic systems.

The formation and evolution of a rectangular synthetic jet (high aspect ratio) was studied by Smith and Glezer [1]. During the first half of the cycle (forward stroke), the flow separates at the orifice exit edges, subsequently producing a pair of vortices. These vortices travel away from the orifice, due to their own induced velocity, and are slightly affected by the fluid suction generated at the second part of the cycle. The streamwise velocity magnitude peaks, at a given downstream location, as the vortex pair passes that region.

Synthetic and continuous jets have been compared experimentally for similar Reynolds numbers using time averaged data [2]. The normalized velocity profiles collapsed for all cases studied ($695 < Re < 2200$). Similar to the continuous jet, the synthetic jet width grows linearly with respect to the downstream direction, although it is wider than the continuous case. The volume flux also has a dissimilar behavior, given that the vortex pair entrains more fluid into the synthetic jet, compared to the continuous jet .

Several studies have been conducted on the impingement of a synthetic jet upon a heated surface [3-9]. An optimum jet-to-surface distance has been identified that leads to maximum local Nusselt numbers [3-5, 7-9]. For separation distances that place the synthetic jet in the intermediate-flow regime, the increased jet velocity coupled with a more effective mixing with ambient air improves local heat transfer rates. Moreover, it has been experimentally observed that the heat transfer is increased when the jet frequency is close to the resonance frequency of the driver cavity [3-6].

The effect of the jet formation frequency and and Reynolds number was studied by Pavlova and Amitay [5]. They found that synthetic jets removed heat up to three times better than the continuous case. Furthermore, for small jet-to-surface distances, higher frequencies presented better heat transfer, whereas for larger distances, lower frequencies are more effective.

Valiorgue et al. [9] found that, for a fixed jet-to-surface spacing in a round jet, two different flow regimes can be identified. For large stroke lengths at $L/D > 5$, a time-averaged recirculating vortex is established at $r/D = 2$. Since the distance traveled by a propagating vortex scales with the stroke length, it is expected that the ratio of stroke length to jet-to-surface spacing L_0/H determines the flow regime. They also found evidence for a power law relationship between Nu and Re with an exponent $n = 0.32 \pm 0.06$.

Numerical simulations of the synthetic jet with heat transfer have recently been conducted [8, 10]. Utturkar et al. [8] studied the effect on the heat transfer of a synthetic jet as a cross flow and the impingement over a heated surface. They assumed laminar flow throughout, since the high effective viscosity was expected to damp the small vorticity scales in the jet orifices. Fairly good agreement was found between experimental and numerical data for this 2-D problem. The numerical simulations justified that synthetic jets might be similarly effective for cross flows compared to impinging flows.

Wang et al. [10] performed the numerical simulation of a cross flow synthetic jet over a heated zone. The vibration of the diaphragm used to produce the pulsating flow was simulated as a sine wave velocity function at the inlet boundary. The model selected to represent the complex convective problem was the large eddy simulation, showing acceptable agreement with experiments for velocities and temperature distributions.

In the majority of the previous work on the subject, the synthetic jet was created using an oscillating diaphragm, where the diaphragm is actuated using piezo-resistive devices. The objective of the present work is to de-couple the behavior of the synthetic jet from the manner in which it is created. If successful, the resulting problem will allow the investigation of fundamental behavior that is not influenced by the design of the device. As such, we postulate a canonical problem in which the oscillating jet is created in a frictionless 2-dimensional parallel planes channel supplied by a sinusoidally oscillating flow. A Finite Volume numerical method was used to perform numerical experiments on the canonical synthetic jet impinging over a heated surface. The simulations were used to investigate (i) the evolution of the velocity and temperature fields throughout the process, (ii) the effect of the Reynolds number Re, jet-to-surface distance H, and the driving frequency f on the heat transfer. The results were compared to those obtained for steady continuous jets and preliminary efforts were made to establish a metric by which to evaluate the cost effectiveness of the cooling approach.

PHYSICAL SITUATION AND MATHEMATICAL MODEL

A canonical geometry was created to study the impingement of a synthetic jet over a heated wall. The fluid was considered to flow between two inviscid, adiabatic parallel plates separated by a distance w. The inlet flow was considered to be an oscillating uniform flow. The canonical problem is shown in Fig. 1

The following assumptions were made in the numerical formulation: (i) Two-dimensional flow, (ii) unsteady laminar flow throughout, (iii) insignificant viscous dissipation, and (iv) thermophysical properties not dependent on temperature. With these approximations, the governing conservation equations for mass, momentum and thermal energy are given as follows:

Continuity:

$$\nabla \cdot \mathbf{v} = 0 \tag{1}$$

Equation of motion:

$$\rho \frac{D\mathbf{v}}{Dt} = -\nabla p + \mu \nabla^2 \mathbf{v} \tag{2}$$

Equation of energy:

$$\rho c_p \frac{D\theta}{Dt} = k \nabla^2 \theta \tag{3}$$

In order to simulate the injection and suction of fluid, the uniform velocity at the channel inlet follows a sinusoidal function as follows:

$$v_{in}(t) = v_{max} \sin(2\pi f \cdot t) \tag{4}$$

Following the practice generally followed in the literature, the Reynolds number Re is defined in terms of the streamwise velocity averaged over the channel exit ($y = 0$), and the first half of the cycle:

$$Re = \frac{U_0 w}{\nu} \tag{5}$$

Where,

$$U_0 = L_0 f \qquad L_0 = \int_0^{1/2f} v_{in}(t)\, dt \tag{6}$$

L_0 is the stroke length which characterizes the amount of fluid ejected during the blowing part of the cycle ($0 < t/T < 0.5$). This approach has been used successfully to compare continuous and synthetic jets [2].

The pulsating flow impinges over a uniform heat flux wall located at a distance H from the channel exit thereby causing convective heat transfer from the wall to the jet. The heat transfer coefficient h and the Nusselt number Nu are defined as:

$$h = \frac{q''}{\theta_{wall} - \theta_{\infty}} \qquad Nu = \frac{h \cdot w}{k} \tag{7}$$

The pumping power P required to produce the unsteady pulsating flow can be defined by a simple energy balance in the inviscid channel:

$$P(t) = \frac{\dot{m}(t)\Delta p(t)}{\rho} \tag{8}$$

Where Δp is the pressure gradient between the channel inlet and exit, ρ is the fluid density and $\dot{m}$ is the mass flux.

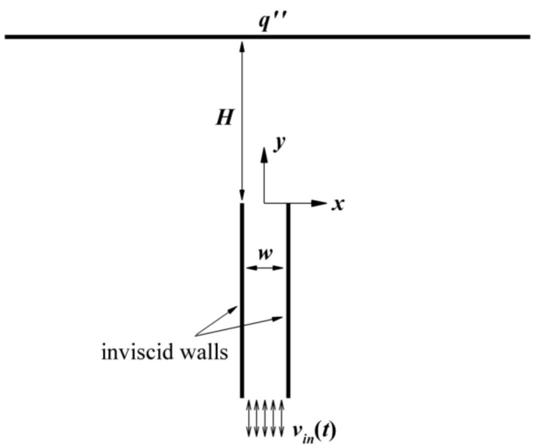

Fig. 1 Physical domain and boundary conditions.

SOLUTION PROCEDURE

The transient mathematical model was solved by the Finite Volume method, by means of the commercial software FLUENT^{TM}. The SIMPLE algorithm was selected to couple velocity and pressure as it performs well for unsteady flows where the time steps are small, as in the present study [11]. First order interpolation was considered for every discretized equation.

The boundary conditions utilized are depicted in Fig. 2, as well as the grid size density where the control volumes with smaller areas are represented by darker gray. For the present study two different types of meshes were used: a structured mesh for the area where the jet develops (y > 0) in order to secure better convergence; and an unstructured mesh for the rest of the computational domain to decrease the number of nodes, taking into account that the fluid is much less dynamic in this region (y < 0).

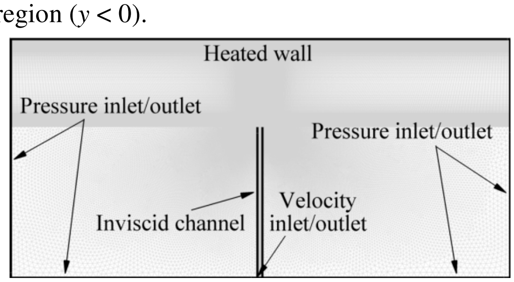

Fig. 2 Computational domain and grid size density.

Mesh size independence and dynamic time step

In order to prove mesh independence on the simulation, four different mesh sizes were tested. The dependence of wall Nusselt number and jet mass flux on grid density is shown in Fig. 3. It can be seen that the mesh with 158,661 nodes introduced only minor deviations for the time averaged Nu, with respect to the finest grid (Fig. 3(a)), a tendency that is also observed in Fig. 3(b), where the two coarser grids showed observable deviations from the theoretical mass flux, leading to a computational error associated with the artificial generation of mass within the system.

Since one of the parameters changed in this study is the jet-to-surface distance H, the size of the computational domain, or the grid size, was varied as well. Thus, in order to maintain the grid independence, the density of nodes per unit length was kept constant in the vertical direction.

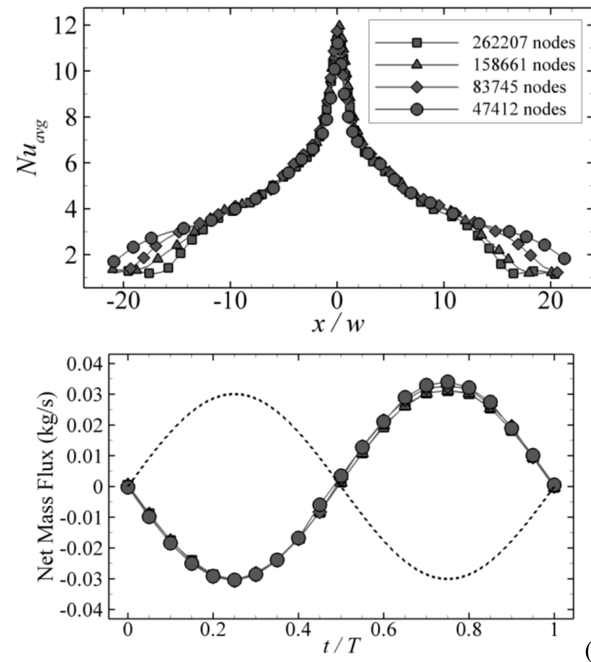

Fig. 3 (a) Time averaged Nusselt number distribution over the heated wall, and (b) Net mass flux at the pressure inlet-outlet boundary compared to inlet condition (dotted line).

Three different time steps, defined as function of the period T, were tested as it is illustrated in Fig. 4. Even though the two cases almost coincided at every location along the heated wall, T/200 was the option elected, due to the fact that this improves the convergence of the simulation, especially in the areas where the velocity gradients are high.

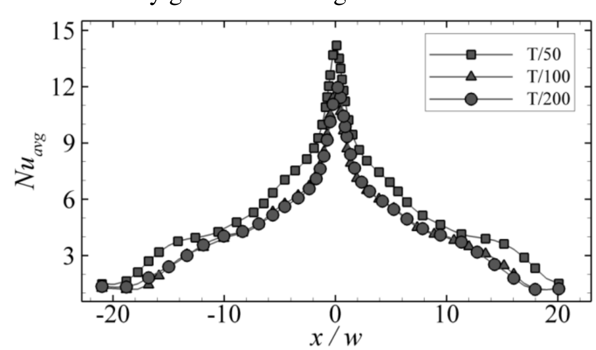

Fig. 4 Time averaged Nusselt number distribution over the heated wall for three different time steps.

RESULTS AND DISCUSSION

The instantaneous velocity field and its corresponding heat transfer response over the heated wall are shown in Fig. 5 for

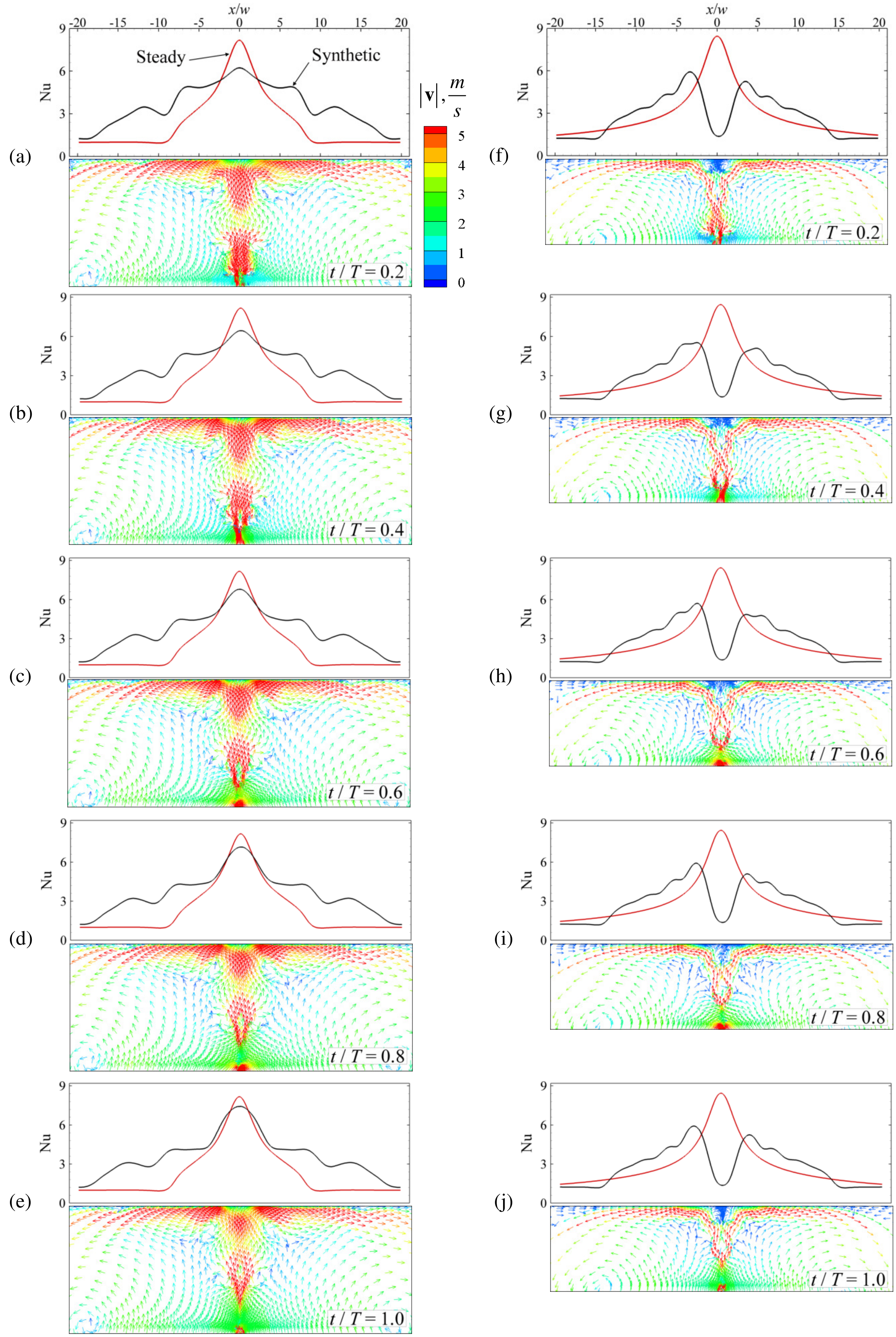

Fig. 5 Instantaneous velocity field colored by the velocity magnitude (bottom) and Nusselt number distribution (top) over the heated wall (Re = 305 and f = 1200) for H/w = 15 ((a)-(e)) and H/w = 10 ((f)-(j)).

one cycle. During the forward stroke ($t/T < 0.5$) a vortex pair is formed at the channel exit edges and subsequently, expulsed in the streamwise direction.

Thereafter, when the suction part of the cycle takes place ($t/T > 0.5$), the vortices have been already sufficiently removed to not be affected by the counterflow, thereby generating non-zero momentum flux in the positive streamwise direction, despite the fact that the net mass flux put into the system is zero. The jet velocity peaks with the passage of the vortex pair at a given downstream location, hence, the Nusselt number increases as the vortices approach the wall. A secondary peak in the distribution of *Nu* is associated with the preceding vortex pair that is advected along the heated wall in the spanwise direction. Because of the highly unsteady nature of the flow, surprising results are evident. For example, in Fig. 5 ((f)-(j)), the vortex dynamics and their interaction with the wall are such that the downwash of fluid toward the stagnation zone is weak, leading to a low velocity zone and a resulting drop in the stagnation Nusselt number.. Hence, depending on the distance to the wall and the Reynolds number, the impinging flow may resemble a well developed impinging jet flow, almost indistinguishable from the steady jet (in the time mean), or a flow dominated by a highly coherent vortex pair.

As previously observed [2], for matched Reynolds numbers, synthetic jets are wider and slower than steady jets. This can be detected in the distribution of *Nu*, where the continuous case reaches a higher maximum; however the synthetic jet removes heat more effectively over a broader area of the heated wall. It can be also observed that, for $Re = 305$, the velocity of the jet remained closer to values of 5 m/s, even though the maximum velocity per cycle at the channel inlet was 15 m/s. In addition, the steady jet velocity for the same Reynolds number was 4.78 m/s, which implies a remarkable resemblance between continuous and pulsating jets, further supporting the definition of *Re* adopted for synthetic jets.

Nusselt numbers, averaged over space and time, as a function of orifice-to-surface distance *H* are shown in the Fig. 6. It can be seen that *Nu* scales monotonically with *Re*, regardless of the frequency at which the fluid is injected into the system. Furthermore, the rollup of the vortex pair improves the entrainment of mass into the jet, augmenting the volume flux, which results in an enhancement of the heat transfer, compared to the steady case, dependent on the jet to wall distance. The improved mixing of the synthetic jet with the external fluid at ambient temperature, as well as a stronger penetration into the thermal boundary layer implies that synthetic jets remove heat in a more efficient manner, compared to steady jets for matched Reynolds numbers.

Commonly, three zones are identified for synthetic jets along the streamwise direction: near-field ($H/w < 7$), intermediate-field ($7 < H/w < 18$) and far-field ($H/w > 18$) [3].

As for steady jets, the heat transfer increases for smaller orifice-to-surface separations ($H/w < 8$), however there is a secondary optimum distance H/w for synthetic jets, in general, located within the intermediate-field range ($12 < H/w < 20$). This phenomenon was more pronounced at lower frequencies, since in that case, a larger stroke length produced larger vortices, thus, enhancing the previously mentioned mixing with ambient fluid.

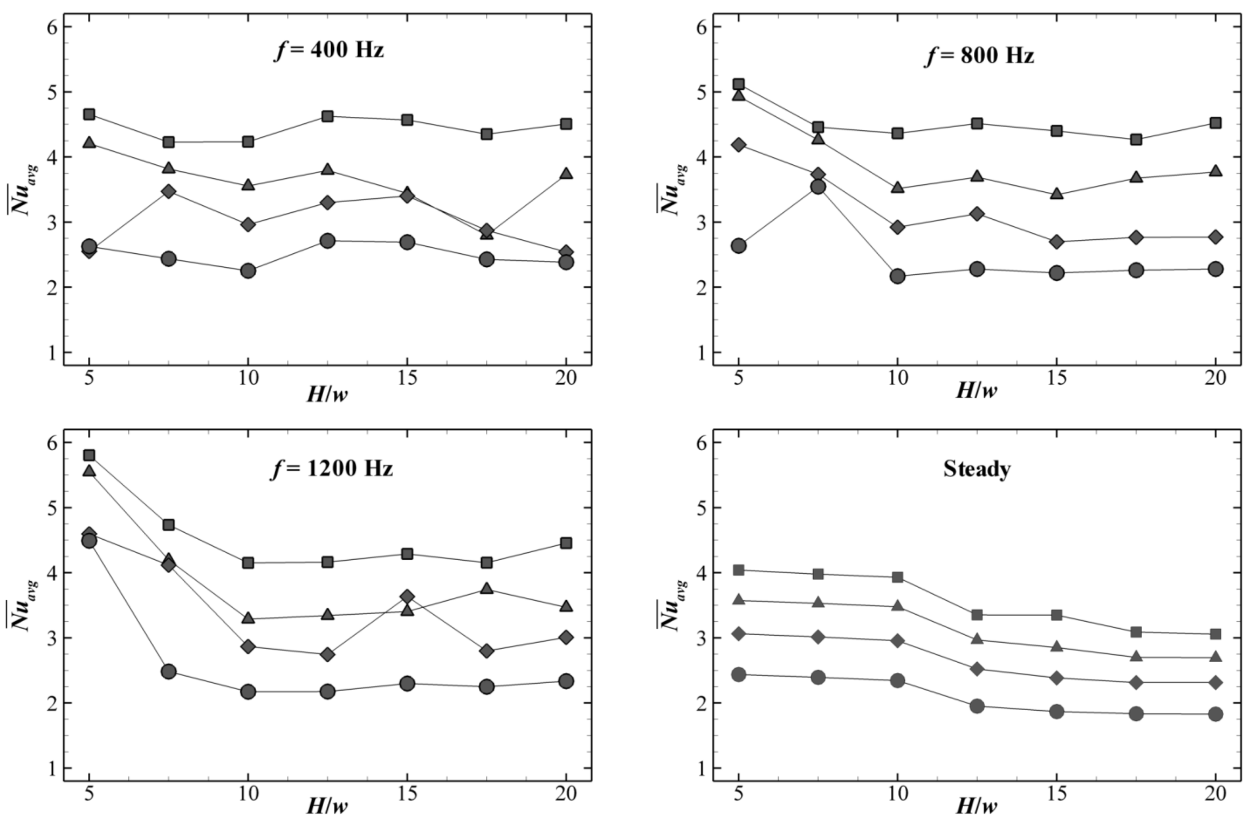

Fig. 6 Space and time averaged Nusselt number for different jet-to-surface distances H/w. (Re = 508(□), 406(□), 305(□), 203(○)).

Time and space averaged *Nu* versus driving frequency *f* is shown in Fig. 7. The frequency affects *Nu* most strongly in the near-field. In the intermediate field there is no evidence of a dependence of the heat transfer on the frequency, in other words, the synthetic jet, for constant *Re*, is not influenced by the way it is produced. As was previously shown by Smith and Glezer [1], the vortex pair breaks down and begins to lose its coherence as it moves downstream. This implies that the synthetic jet diminishes its capacity of mixing with the adjacent fluid for larger values of *H*/*w* (intermediate-field), leading to lower *Nu*. Therefore, at orifice-to-surface distances placed in the near-field, higher driving frequencies generate a more dynamic velocity field, with a subsequent improvement of the heat transfer (Fig. 5(a)).

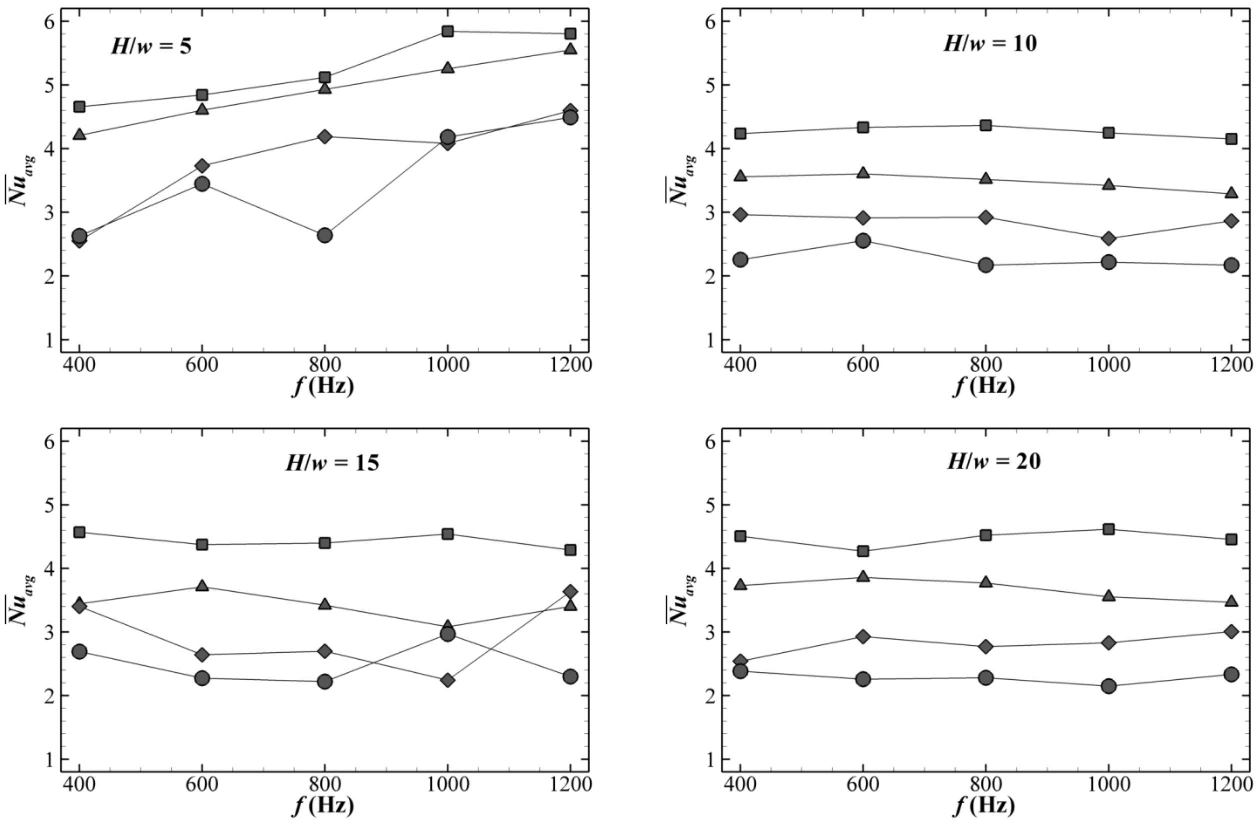

Fig. 7 Space and time averaged Nusselt number as a function of frequency: *Re* = 508(□), 406(□), 305(□), 203(○).

The time response of the stagnation Nusselt number Nu_s normalized with its correspondent averaged $\overline{Nu}_{avg}$, over two cycles, is depicted in Fig. 8. As expected, higher values of *Re* lead to higher amplitude in the oscillations, since the fluid that impinges over the stagnation zone ($x = 0$) is also more affected by the pulsating jet. As has been previously mentioned, the streamwise velocity of the jet maximizes with the passage of the vortex pair, due to the downwash of fluid between the vorticies; hence the instant where Nu_s peaks is associated with the moment where the vortices reach the heated wall. For that reason, different frequencies and jet-to-surface distances result in a shifting of this peak with respect to the instant where the jet velocity reaches its maximum ($t/T = 0.25$). Furthermore, it can be seen that even though the stagnation Nusselt number has a time periodic behavior, the curves are not symmetric between periods. The arrival of the vortex pair to the heated wall rapidly augments the heat transfer up to its maximum; thereafter the stagnation zone is subjected to the impingement of the slower fluid until the next pair repeats the process.

The instantaneous power required to generate the oscillating jet during one cycle is illustrated in Fig. 9. From a thermodynamic energy balance on the channel, the power input into the system goes into increasing the enthalpy and kinetic energy of the exiting fluid. However, the change in internal thermal energy and kinetic energy is negligible from inlet to exit, and hence the power consumed is represented almost entirely by the mechanical flow work component of the enthalpy increase. By increasing the driving frequency the power also increases (Fig. 9(a)), for constant Reynolds number (constant inlet velocity amplitude). This is due to the fact that the fluid within the channel is required to reach the same velocity maximum regardless of *f*, therefore as the frequency increases the time interval to do so diminishes. As a consequence of this, the higher is *f*, the more rapid is the acceleration and deceleration of the fluid, and since the mass flux is also constant for constant *Re*, the power augments proportionally to *f*.

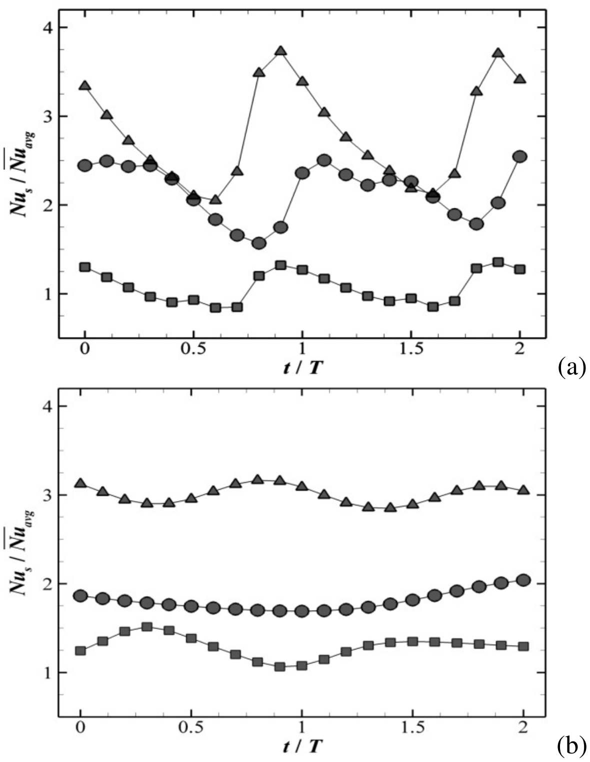

Fig. 8 Instantaneous stagnation Nusselt number for two cycles: (a) Re = 508, f = 400 Hz and (b) Re = 305, f = 1200 Hz (H/w = 5 (□), 10 (□), 15 (○)).

The pumping power is also directly proportional to Re, as is shown in Fig. 9(b). For constant driving frequency, the rising of P is associated with the augmentation on the mass flux when the Reynolds number is increased.

During the pumping process, four stages can be identified through a single cycle, each with duration of $\Delta t/T = 0.25$. For the first part ($0 < t/T \leq 0.25$), the power reaches its first maximum due to the increasing of the velocity v_{in}, and a consequently raising on the mass flux entering the system. Thereafter ($t/T = 0.25$), P goes to zero due to a zero pressure gradient since there is no fluid acceleration ($dv_{in}/dt = 0$). When $t/T = 0.5$ no fluid is being pumped, since $v_{in} = 0$, therefore $\dot{m} = 0$. This process repeats during $0.5 < t/T \leq 1.0$, when the suction part takes place ($\dot{m} < 0$). In essence, during one cycle a competition exists between the pressure gradient and the mass flux, as one quantity maximizes when the other approaches zero.

In order to investigate the energy effectiveness of a system designed around the principles of a synthetic jet, we introduce a cost-effectivness coefficient that weights both the efficacy of heat transfer and the power consumed to produce the flow. The cost-effectiveness coefficient is defined as:

$$\eta = \frac{1}{R \cdot P_{avg}} \tag{9}$$

Where $R = (\theta_{wall} - \theta_{\infty})/q''A$ (thermal resistance) and P_{avg}, corresponds to the time averaged pumping power.

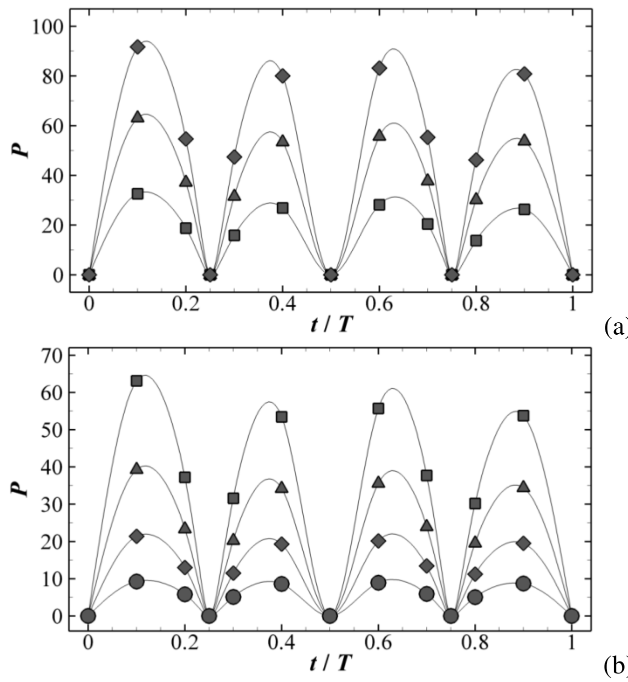

Fig. 9 Instantaneous pumping power P (W) during one cylcle: (a) f = 400(□), 800(□), 1200(□) and (b) Re = 508(□), 406(□), 305(□), 203(○).

The cost effectiveness as a function of H for different Re is shown in Fig. 10 (f = 800). From the stand point of η an optimum jet-to-surface distance can be identified (H/w = 5) for a fixed driving frequency. The pumping power remains essentially constant, for the same Re, regardless of H, hence the better heat removal due to the proximity of the heated wall to the jet orifice yields to a higher effectiveness. Moreover, it can be seen that the lower is Re the higher is η. For instance, by increasing the Reynolds number from 203 to 508, at H/w = 5, the pumping power augmented approximately 660%, whereas Nu was enhanced nearly by a 100%.

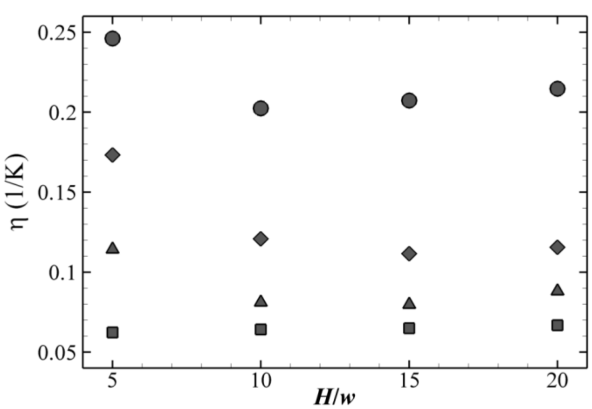

Fig. 10 Cost-effectiveness versus jet-to-surface distance H/w for f = 800 Hz (Re = 508(□), 406(□), 305(□), 203(○)).

CONCLUSIONS

The simulation of a synthetic jet impinging on an evenly heated wall was presented. A numerical solution of the discretized system of equations was performed by the finite volume method using the SIMPLE algorithm. The thermophysical properties were considered constant throughout. Some general conclusions can be drawn from the study as follows:

1. For matched Reynolds numbers, synthetic jets are slower and wider than the steady case; therefore the heat is removed more efficiently across the same impinging area, even though, for some cases, in the vicinity of the stagnation zone *Nu* was higher for continuous jets.
2. Similarly to steady jets, the Nusselt number increased: (i) as *Re* was augmented due to a higher convection of fluid towards the heated wall and (ii) when H/w was diminished as the jet impinged with higher momentum when the nozzle is closer to the wall. Nevertheless, for the synthetic jet a secondary optimum was observed located in the intermediate-field ($12 < H/w < 20$), associated with a more effective mixing with air at ambient temperature.
3. An influence of the driving frequency upon the heat transfer was only observed in the near-field ($H/w < 10$), whereas for the intermediate-field, *Nu* showed no dependence on the way that the jet was generated. It is hypothesized that as the vortex pair moves towards the wall, it breaks down and its coherence diminishes, therefore in the near-field, the heat transfer phenomenon is dominated by the rotation of the vortices, as opposed to the impinging of a quasi-steady synthetic jet that was observed for larger H/w.
4. Heat was removed more effectively in the near-field by increasing f, whereas in the intermediate-field this was achieved by diminishing the frequency.
5. The pumping power P required to generate the synthetic jet was investigated, showing to be in direct proportion to *Re*, since a higher inlet velocity led to a higher mass flux added and removed per cycle. The pumping power increased as f was raised (for constant *Re*), since the fluid was required to accelerate up to v_{max} and decelerate down to $-v_{max}$ in a shorter period of time. Nevertheless, for the range $5 < H/w < 20$, P presented no variation with respect to the jet-to-surface distance.
6. A cost-effectiveness ratio between the pumping power and the removed heat was defined in order to estimate the efficiency of the synthetic jet. It was found that the cost effectiveness coefficient was highest for shorter H/w and lower Reynolds numbers. The implication that the synthetic jet is most efficient at heat removal flow low Reynolds numbers and close jet to target spacing.

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